

Spring Boot and Microservices Syllabus

Archer Infotech, Pune | /courses/programming/spring-boot-microservices-training-in-pune

Duration	10 to 12 weeks plus production capstone
Level	Java foundation to backend microservices
Format	Classroom, online live and weekend-friendly batches
PDF purpose	Downloadable syllabus for website visitors, counsellors and enrolled learners

Clear Scope Analysis

This PDF is the follow-on Java backend track. It assumes Core Java comfort and moves into Spring Boot APIs, persistence, security, microservices, messaging, testing, observability and deployment.

The sequence is designed for learners who already know Java basics and need a portfolio-grade backend project suitable for Pune Java developer interviews.

What You Will Learn

- Spring Boot Foundations
- REST API Development
- Spring Data JPA and Databases
- Security and Authentication
- Testing Spring Applications
- Microservices Architecture
- Messaging, Observability and Deployment
- Production Capstone

Detailed Module Sequence

Module 1: Spring Boot Foundations

Create Spring Boot projects, understand starters, auto-configuration, application properties, profiles and layered architecture.

Module 2: REST API Development

Build controllers, DTOs, validation, exception handling, pagination, filtering, OpenAPI documentation and clean API responses.

Module 3: Spring Data JPA and Databases

Use entities, repositories, relationships, JPQL, transactions, migrations, indexing basics and query performance habits.

Module 4: Security and Authentication

Implement password hashing, JWT login, roles, method security, CORS, secure configuration and practical API protection.

Module 5: Testing Spring Applications

Write unit tests, slice tests, integration tests, Testcontainers awareness, mocks and API contract verification.

Module 6: Microservices Architecture

Split services, define boundaries, use service discovery, API gateway concepts, resilience, retries and central configuration.

Module 7: Messaging, Observability and Deployment

Use Kafka/RabbitMQ concepts, logs, metrics, traces, Docker, CI/CD and cloud deployment readiness.

Module 8: Production Capstone

Build a multi-service Spring Boot backend with database, security, tests, documentation and interview-ready architecture notes.

Projects

- Secure REST API with database
- Order and inventory microservices
- Production-style Spring Boot capstone

Learner Outcomes

- Spring Boot Developer readiness
- Java Backend Developer portfolio
- Microservices interview preparation

Download Note

This syllabus has been regenerated from the corrected Archer Infotech programming course content with SEO, GEO, AEO and AI-understanding friendly structure: clear title, explicit scope, sequenced modules, descriptive outcomes and unambiguous follow-on boundaries.